INTERACTION OF POLYDISPERSE DROPLETS WITH AN AIR VORTEX RING: TOWARDS UNDERSTANDING ATMOSPHERIC CLOUD

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Introduction:

Atmospheric clouds regulate the hydrological cycle and Earth's radiation budget by absorbing long-wave radiation and reflecting short-wave solar radiation. In **warm clouds**, this influence is closely linked to the growth of droplets into rain. The process begins with the nucleation of cloud condensation nuclei ($\sim 1\ \mu m$), followed by condensational growth to droplets ($\sim 10\ \mu m$), and further growth toward drizzle sizes ($\sim 100\ \mu m$), after which droplets begin to fall under gravity. Raindrops ($\sim 1\ mm$) form from cloud droplets ($\sim 100\ \mu m$) primarily through gravitational collision and coalescence [1], as schematically illustrated in Figure 1. The transition from $10\ \mu m$ to $100\ \mu m$, commonly referred to as the "bottleneck problem," remains poorly understood [2-3]. Turbulence is widely hypothesized to accelerate this growth. **The present study investigates droplet interactions with controlled vortical structures to quantify droplet redistribution, clustering, and collision/coalescence pathways relevant to this bottleneck.**

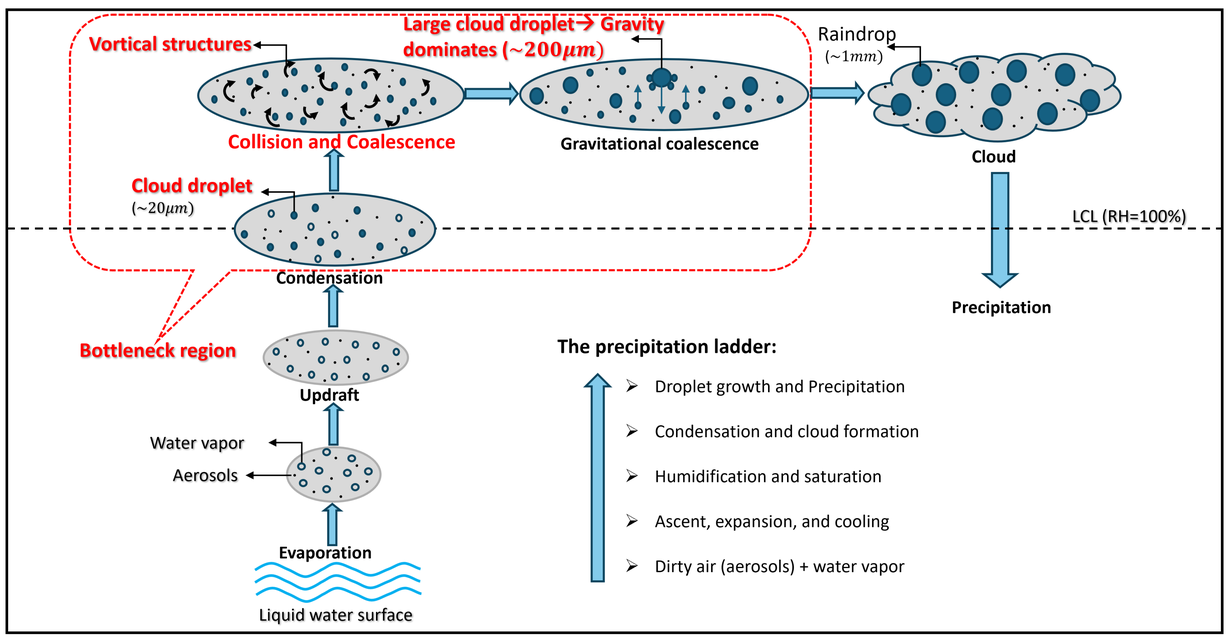

Figure 1: Schematic of cloud formation and rain initiation

Approaches to address the bottleneck problem:

Multiphase DNS	Multiphase LES	Experiments
➢ $\frac{l}{\eta} = 10^5$ ➢ No. of grid points • For 2D domain : 10^{10} • For 3D domain : 10^{15} **Computationally challenging!**	➢ Useful to understand the cloud dynamics. ➢ Cannot resolve smallest scales of turbulence and the cloud droplet scales. **All scales are not resolved!**	➢ Can resolve the scales associated with the cloud microphysics. ➢ Comparatively easy to track the droplets using flow diagnostic techniques like PIV, PTV.

Experimental methodology:

Clouds may be idealized as turbulent regions of air and water vapor containing eddies over a wide range of scales that promote droplet growth through collision and coalescence. However, the inherent complexity of real clouds makes it difficult to isolate the mechanisms governing droplet–vorticity interactions, clustering, and collision–coalescence processes. To address this, the eddies are idealized as localized vortical structure, a vortex ring [4], enabling a controlled investigation of vortex–droplet interactions. In particular, a single Gaussian vortex ring is generated to examine droplet dynamics within an isolated vorticity field. Subsequently, to approach more realistic atmospheric conditions, the study is extended to a turbulent boundary layer (TBL), where multiple vortical structures coexist and interact.

The experimental facility is shown in Figure 2. A controlled Gaussian vortex ring is generated and characterized using TR-PIV at 4000 Hz, with ~1 μm fog particles as tracers. The vortex ring (Figure 3a) has a core diameter of 2.7 cm and a circulation strength of 5080 cm²/s.

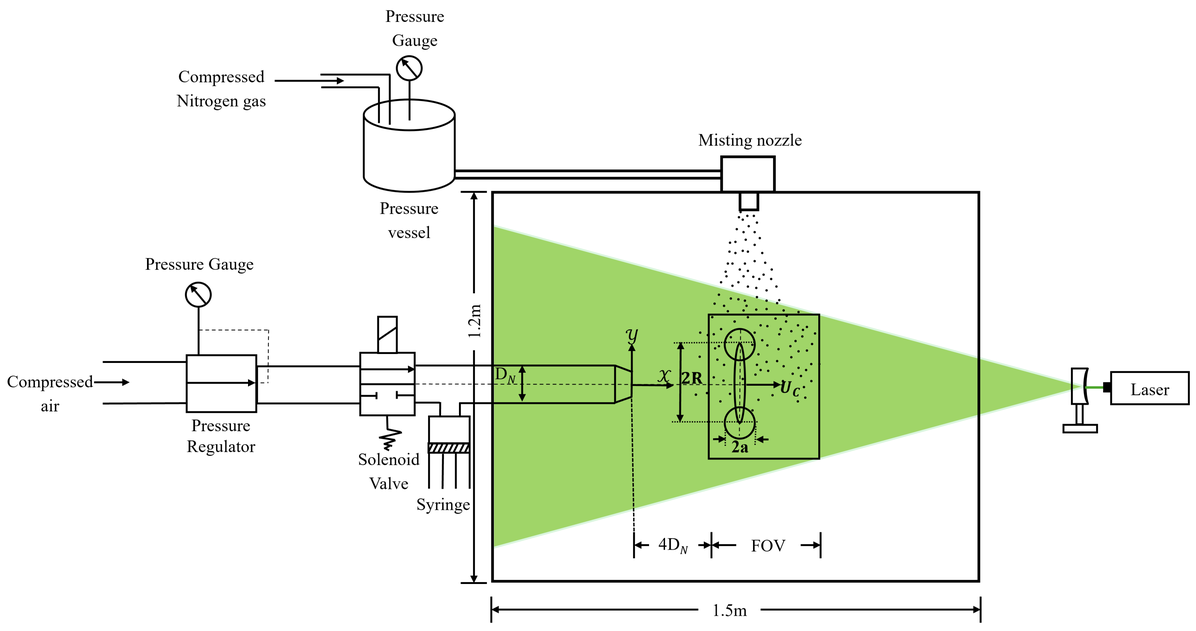

Figure 2: Schematic of experimental facility available at IIT Delhi

Polydisperse droplets (50–250 μm), produced using pressurized spray nozzles (Figure 3b), are introduced to examine vortex-induced redistribution and preferential concentration, influencing collision and coalescence within the flow field (Figure 3c).

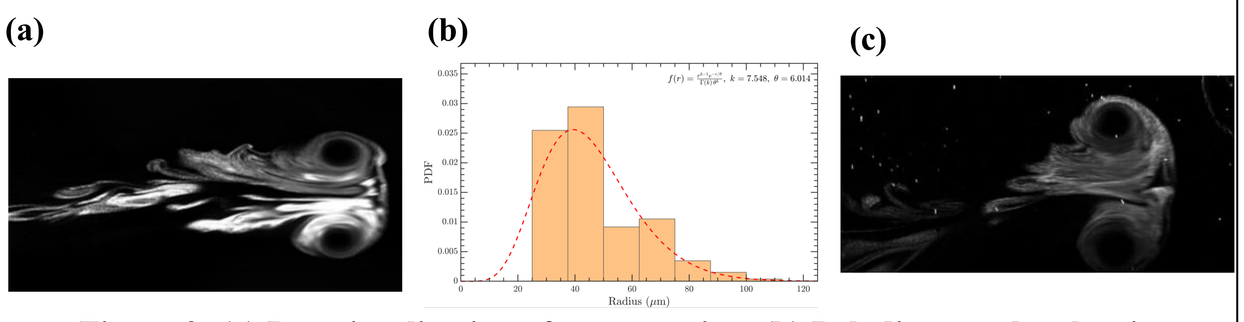

Figure 3: (a) Dye visualization of a vortex ring, (b) Polydisperse droplet size distribution (c) Interaction of multiple droplets with a Vortex ring

Results and discussion:

The vortex-ring–droplet interaction induces strong droplet redistribution: droplets migrate from high- to low-vorticity regions, forming clusters and voids (Fig. 3a,b). To quantify this spatial organization, Voronoï analysis is employed and compared with a Random Poisson Process (RPP) representing a uniform distribution [5] (Fig. 3c). The experimental PDF deviates clearly from the RPP case, confirming vortex-induced preferential concentration (Fig. 3c). The degree of clustering increases as the vortex propagates, with a mean relative clustering of $\sigma_{rel} = 0.74$. Such preferential concentration can promote caustic formation and enhance droplet collision and coalescence [6].

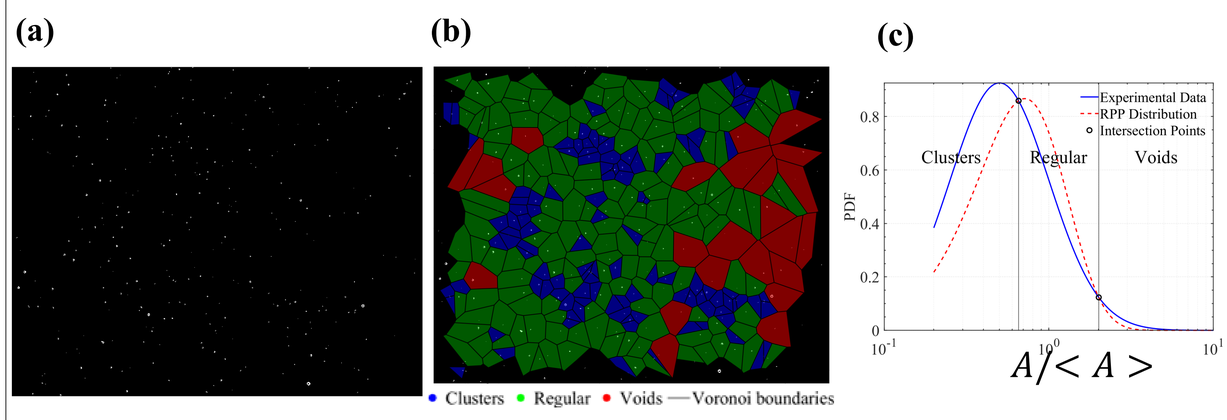

Figure 3: (a) Binary image of the polydisperse droplets, (b) Voronoi diagram showing the clusters and voids generated after the vortex influences the Droplets ; (c) Comparison of pdfs of experimental normalized Voronoi area with respect to the random Poisson process (RPP) distribution.

Conclusion:

We experimentally examined the interaction of polydisperse droplets (50–250 μm) with a single Gaussian vortex ring. The interaction induces preferential concentration, forming clusters and voids, as confirmed by Voronoï analysis. This vortex-driven organization may promote caustic formation and enhance collision and coalescence relevant to cloud droplet growth.

References:

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